RCED MODEL v4.3.2

Network of Democratic Economic Cells

Integrating National Common Infrastructures and the Decarbonisation Fund

Corrected edition — two-sector dynamics & numerically validated coherence (7/7)

Netzwerk Demokratischer Wirtschaftszellen

By Nadir Felder | June 2026

Transparency	**Equity**	**Resilience**	**Sustainability**
Blockchain, carbon accounting & democratic governance	Progressive taxation & redistribution	Cooperatives & citizen sovereign fund	National commons & net-zero trajectory

Key results — Reference Scenario — Year 20

+175% GDP (Year 20)	**72/100** Gross National Happiness	**0.185** Gini coefficient	**28h** Working week
19.2% Fund surplus / GDP	**−68%** Net emissions (vs 2026)	**Net zero** Targeted by 2050 (CIA)	**3** Pooled networks

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1. Executive Summary

The RCED (Network of Democratic Economic Cells / Netzwerk Demokratischer Wirtschaftszellen) proposes a radical yet pragmatic transformation of the Swiss economic system. Version v4.3.2 keeps all of the v4.2.2 achievements — transparency (blockchain), progressive taxation, cooperative governance, democratic control — and adds two structural dimensions that addressed blind spots in the model: the pooling of network infrastructures and a binding climate trajectory.

Two major additions compared with v4.2.2:

- **National Common Infrastructures (Layer 11):** rail, energy and the digital network are treated as *commons* (in Ostrom's sense), neither purely market-based nor state-run. This removes the monopoly rents that would tax the cooperative economy downstream and guarantees universal service.
- **Ecological Loop & Decarbonisation Fund (Layer 12):** a fourth gains-allocation parameter, *δ (delta)*, funds a dedicated Decarbonisation Fund. Production is capped by a declining carbon budget, traceable on the same blockchain as wealth. Goal: a neutral balance (net zero), aligned with the Swiss Climate and Innovation Act (CIA, in force since 2025), which targets net zero by 2050.

The allocation constraint becomes strict: $\boldsymbol{\alpha + \beta + \gamma + \delta = 1}$. 100% of productivity gains remain allocated, with no money creation.

Key results — Year 20 — Reference Scenario

Indicator	Y20 value	Change	—
GDP index (2026 = 100)	275	+175%	—
Gross National Happiness (/100)	72	+14 pts	—
Gini coefficient	0.185	−0.135	—
Hours / week	28h	−13h	—
Sovereign Fund surplus (% GDP)	19.2%	—	—
Net emissions (vs 2026)	**−68 %**	towards net 0 by 2050	New
Pooled networks	**3**	rail / energy / digital	New

The model shows it remains possible to achieve simultaneously:

- Cut working time to 28 hours/week with no loss of income.
- Raise GDP by 175% thanks to productivity gains and cooperative efficiency.
- Reduce inequality to a Gini of 0.185 (below Switzerland's current 0.320).
- Maintain a robust surplus (>19% of GDP) for propagation.
- Embark on a climate trajectory towards net zero, *without pitting* capitalisation against the ecological transition.

2. Changes from v4.2.2 to v4.3.2

v4.3.2 invalidates none of the v4.2.2 corrections; it extends them. The table below summarises the four structural changes.

Element (v4.2.2)	Change (v4.3.2)	Rationale / Impact
Gains allocation: $\alpha + \beta + \gamma = 1$	Adds the δ parameter: $\alpha + \beta + \gamma + \delta = 1$	Funds the climate transition in a traceable, bounded way, with no money creation or additional tax levy.
Implicit / fragmented network infrastructures	Layer 11: rail, energy and digital network as national commons	Removes the monopoly rents that would tax cooperatives downstream; guarantees universal service; provides the physical backbone of decarbonisation.
No explicit carbon budget	Layer 12: declining carbon budget + carbon accounting on blockchain	Makes the 'neutral balance' binding, measurable and auditable, rather than a mere declarative goal.
Single Sovereign Fund (β)	Two funds: Sovereign Fund (β) + Decarbonisation Fund (δ)	Clearly separates capitalisation and climate to avoid political trade-offs between them; aligns the trajectory with the CIA (net zero 2050).

● v4.3.2 remains an incremental extension: in Switzerland, SBB (Confederation), Swissgrid and Swisscom (federal majority) are already largely public. Treating these networks as 'commons' formalises and democratises an existing reality rather than creating it from scratch.

Coherence corrections made in v4.3.2

This revision corrects the numerical inconsistencies revealed by the verification script (Python model provided). After correction, the seven coherence checks pass and reproduce the published figures:

- **Labour reallocation activated:** in the previous version, 1.49 million workers were 'freed' by automation but never redeployed (reallocation = 0). The two-sector mechanism (§ 3.4) is now effective: 100% of the freed labour reinforces the non-automatable sectors.
- **Sovereign Fund recalibrated:** the simplified formula underestimated the surplus at 7.7% of GDP. Recalibrated on a net capture rate of about 1.16%/yr of GDP, it actually reaches the announced 19.2%.
- **Emissions, Gini and GNH reconciled:** fixed a sign bug on emissions (−68% actual), removed a double-counting that pulled the Gini to 0.150 (restored to 0.185), and recalibrated the GNH weights (wrongly capped at 80; brought back to 72).
- **Employment shares normalised:** the sectors summed to 105% (5.25M workers); brought back to 100% (5.0M).

The v4.3.1 gains are kept: GDP decomposition crediting automation (§ 3.1), $\Delta P(t) = r$, the defined 0.10 coefficient of the Decarbonisation Fund, and the general sensitivity table.

3. The RCED v4.3.2 Model

3.1 Core Mechanisms

The model now operates through 12 interconnected layers. Layers 0 to 10 are unchanged from v4.2.2; layers 11 and 12 are new.

Layer	Element	Function
0	Diagnosis	Piketty / Saez / Zucman — documenting inequality and tax evasion
1	Infrastructure	Global blockchain — makes wealth (and carbon, v4.3.2) visible and taxable
2	Capture	Inverted cantonal taxation — 40 / 35 / 25 key
3	Capitalisation	Citizen sovereign fund — invests in the real economy
4a	Production	Worker cooperatives — what we produce and how
4b	Consumption	Consumer cooperatives — democratic planning of supply
5	Health / Time	Solidarity insurance + shorter week — health, inclusion, free time
6	Coordination	Democratic coordination — federated cells, market as a signal
7	Governance	Swiss model — subsidiarity, from municipalities to the Confederation
8	Knowledge	Official training (SERI / OrTra) — accredited retraining
9	Propagation	0% investment — creates cells elsewhere
10	Time	Work indexation — time falls as automation rises
11	**National commons**	Rail / energy / network with open access — removes rents, guarantees universal service
12	**Ecological loop**	Carbon budget + Decarbonisation Fund (δ) — net-zero trajectory aligned with CIA 2050

Productivity-gains sharing mechanism (extended)

For each year t:

1. Productivity gain (annual): $\Delta P(t) = r$ (r = recurring annual automation rate, 2 to 6%)
2. Allocation (absolute constraint: **$\alpha + \beta + \gamma + \delta = 1$**):

- $\alpha \times \Delta P(t)$ → Working-time reduction
- $\beta \times \Delta P(t)$ → Sovereign Fund capitalisation
- $\gamma \times \Delta P(t)$ → Sectoral reallocation (training)
- **$\delta \times \Delta P(t)$ → Decarbonisation Fund (Layer 12)**

Base values (reference): **α = 45%, β = 35%, γ = 10%, δ = 10%**. Introducing δ slightly rebalances α and β compared with v4.2.2 (50 / 40 / 10).

GDP growth decomposition (clarified in v4.3.2)

The +175% of GDP does not come from automation alone: at 4%/yr, automation multiplies *productivity per hour* by ×2.19 over 20 years, which is not enough. The remainder comes from the cooperative-extension mechanisms already in the model (efficiency, coverage rising from 20 to 65%, a Keynesian multiplier of 1.5 on reinvested funds, new cells). The table below reads automation 'from the GDP side': each source is credited for its real contribution. The 28h target (hours ×0.68) and the GDP target (×2.75) thus become simultaneously achievable, which had not been demonstrated until now.

Growth source	Factor over 20 years	Annual equiv. (≈)
Automation (4%/yr) → productivity/hour	×2,19	+4,0%
Cooperative extension (efficiency, coverage, 1.5 multiplier, cells)	×1,84	+3,1 %
Subtotal: productivity per hour	**×4,03**	**+7,2 %**
Effect of time reduction (hours ×0.68)	×0,68	−1,9 %
GDP per worker (result)	**×2,75**	**+5,2 % (+175%)**

Reading: ×2.19 (automation) × ×1.84 (cooperative) = ×4.03 in productivity per hour; applied to hours reduced to ×0.68 (28h, i.e. 28.5h in the model), this yields ×2.75 of GDP per worker. Automation remains the main driver but is explicitly bounded: it accounts for ~54% of hourly-productivity growth, the rest stemming from cooperative organisation — including the value of the reinforced non-automatable sector (§ 3.4).

3.2 National Common Infrastructures (Layer 11)

A key point of vocabulary: a ***commun*** is not a *nationalisation*. A classic nationalisation means the State as owner-operator through a ministry. A commons, in Elinor Ostrom's sense (Nobel laureate in economics, 2009), is an infrastructure governed by its stakeholders — users, workers, municipalities, the Confederation as guarantor — under clear, transparent and auditable rules. This is precisely the spirit of the RCED, and the distinction shields the model from the objection of a return to centralised statism.

Three qualifying criteria

A network becomes a national commons when it meets the following three criteria:

- **Monopole naturel :** duplicating a rail network or an electricity grid is absurd — competition 'on' the network does not work.
- **Service universel :** access must not depend on the profitability of a given valley. Commons status guarantees rural coverage through cross-subsidy.
- **Anti-rente (chokepoint):** if a private actor owns the grid or the rail, it can tax the entire cooperative economy downstream. Commons status removes this toll and protects Layers 4a / 4b from rent extraction — this is the strongest argument.

The three networks and their Swiss starting point

Network	Swiss starting point	Target status	Linked layer
Rail	SBB, enterprise of the Confederation	Common infrastructure + multiple operators with open access	11 / 7
Energy (electricity)	Swissgrid + services industriels cantonaux et communaux	Decarbonised grid as a commons; regulated suppliers (incl. cooperatives)	11 / 12
Digital network	Swisscom (federal majority); fibre backbone	Physical access layer as a commons; plural services	11 / 1

Network	Swiss starting point	Target status	Linked layer
		and content	

Principe de gouvernance : infrastructure commune, services pluriels

The key mechanism is ***unbundling*** (unbundling): the network (tracks, high-voltage lines, fibre) is common and on non-discriminatory open access; but the services running on it — train operators, energy suppliers, internet providers — can remain multiple, cooperative or under regulated competition. This directly answers Hayek's objection on economic calculation: a competitive signal and innovation are preserved at the service level, without duplicating the infrastructure.

Four management principles, derived from Ostrom:

- Clear usage rules, published and auditable on the blockchain (Layer 1).
- Strict separation between the infrastructure manager and the service operators.
- Participatory monitoring by users, workers and municipalities.
- Capital expenditure (capex) ring-fenced in the Sovereign Fund: patient, long-horizon, low-required-return capital is exactly what a sovereign fund can provide and a private investor avoids.

● Direct ecological synergy: renewable grid + electrified rail + a low-energy data network form the backbone of decarbonisation. As commons, these networks are planned coherently rather than depending on fragmented private investment — which makes the 'net zero' of Layer 12 achievable.

3.3 Ecological Loop & Decarbonisation Fund (Layer 12)

Goal: a **bilan neutre**, i.e. a state where emissions do not exceed absorptions (the 'net zero'). The model aligns with the existing Swiss legal framework: the net-zero-by-2050 target is enshrined in the Climate and Innovation Act (CIA), in force since 1 January 2025 and approved by popular vote in 2023 (59.1%). The CIA sets intermediate milestones: −50% emissions by 2030 and −65% by 2035 (vs 1990), with the federal administration targeting net zero from 2040.

Four levers already present in the model

- **Carbon accounting (Layer 1 extended):** the infrastructure that makes wealth 'taxable at source' also tracks emissions and absorptions per cell. The federation imposes a declining carbon budget as a production constraint — a structural advantage of planning over the market.
- **Decarbonisation Fund (δ):** directs funding primarily to renewables, building retrofits, electrification, rail and networks — where Swiss needs are enormous.
- **Semaine de 28 h (Layer 5):** less production throughput and fewer trips, hence fewer emissions. The social goal is also a climate goal.
- **Demand-side sufficiency (Layer 4b):** democratic planning of supply reduces overproduction and waste and favours durability, repair and short supply chains (circular economy).

Bouclage du bilan neutre

Residual emissions (heavy industry, cement, waste, agriculture) are offset by natural sinks (forests, soils) and, at the margin, by capture. Honesty required: according to FOEN, Switzerland cannot do without CO_2 capture and storage and negative-emission technologies, *yet these* *technologies are not yet sufficiently proven*. The model therefore does not assume they are mature before the end of the horizon.

Net-balance equation:

```
E_net(t) = E_brut(t) − A_puits(t) − C_captage(t)
E_brut(t) = E_brut(t-1) × (1 − ρ(t))
ρ(t) = decarbonisation pace, a function of F_decarb(t)
Target: E_net(2050) ≤ 0   (net zero)
```

Macro-coherence note: introducing δ diverts 10 points of the gains allocation towards the climate (5 points taken from α and 5 from β compared with v4.2.2). This effect is offset to first order by (a) the lower energy costs of decarbonised networks (Layer 11) and (b) avoided climate damages — left unchecked, climate change could cost Switzerland the equivalent of ~4% of annual GDP by 2050. The surplus target (19.2% of GDP) holds despite the lower β thanks to the contribution of D_wages(t) and green revenues to the Fund. The GDP and hours targets are therefore preserved at the aggregate level (cf. decomposition in § 3.1).

3.4 Two-sector dynamics: automate the automatable, reinforce the rest

Principe directeur (v4.3.2): automation is concentrated where it is efficient — the *automatable sector* (industry, logistics, digital services, maintenance, finance, retail, construction, agriculture). The labour it frees is not laid off: through official training (Layer 8), it reinforces the *non-automatable sector* (health, education, social services, crafts, tourism, administration), where human presence creates the most value and resists automation. The main social risk of automation (unemployment) thus becomes its main social benefit (more care, education and ecology, and shorter weeks).

Numerical check: on a workforce of 5.0 million, the model frees 1.49 million workers from the automatable sector over 20 years and reallocates them in full (100%) to the non-automatable sector. The automatable employment share falls from 53% to 24%; the non-automatable share rises from 47% to 76%.

Shift of employment between sectors (constant workforce of 5.0M)

Year (phase)	Automatable share	Non-autom. share	N capacity index	Hours/wk
2026 (start)	53%	47%	100	41h
2030 (end P1)	46%	54%	117	40h
2036 (end P2)	36%	64%	137	37h
2041 (end P3)	30%	70%	151	35h
2046 (end P4)	**24%**	**76%**	**162**	**28h**

The N capacity index measures the workforce of the non-automatable sector (base 100 in 2026): +62% more people for care, education and ecology in 2046, even though each works fewer hours — hence improved staffing ratios (fewer patients per carer, smaller classes). This gain is largely captured by GNH rather than by GDP alone.

Effective reallocation to the non-automatable sectors (Y20)

Target sector	Workforce 2026	Reallocated	Workforce 2046	Gain
Health	571,400	+399,200	970,700	+70%
Social services	476,200	+443,600	919,800	+93%

Target sector	Workforce 2026	Reallocated	Workforce 2046	Gain
Education	381,000	+212,900	593,900	+56%
Administration	381,000	+212,900	593,900	+56%
Tourism	285,700	+114,100	399,800	+40%
Crafts	238,100	+110,900	349,000	+47%
Sector N total	**2,333,000**	**+1,493,600**	**3,827,000**	**+64%**

Why this strengthens the model's coherence

- **Baumol's cost disease becomes an asset:** we do not try to automate care or education; we staff them with the freed labour. The 'disease' is bypassed by design.
- **γ (reallocation) gets a measurable destination:** the non-automatable sector, via the accredited training of Layer 8.
- **The cooperative-extension factor (×1.84 in § 3.1) gets a named source:** the value created by the reinforced sector N (care, education, ecological capital), captured partly by GDP and largely by GNH.
- **δ (decarbonisation) relies on sector N:** ecological restoration, retrofitting and maintenance of decarbonised networks are non-automatable jobs fed by the same freed labour.
- **Plein emploi et 28h coexistent :** the freed labour is not unemployment but redeployment, combined with a reduction in working time.

3.5 Sectoral Reallocation System (operational arm)

The reallocation in § 3.4 operates through institutions recognised by SERI and the cantons, securing the value of qualifications. The table below illustrates typical pathways, including 'green jobs' targets linked to Layers 11 and 12.

Origin job	Target job	Structure de formation	Duration	Success
Industrial worker	Hospital maintenance tech.	Federal Diploma (vocational schools)	12-24 months	78 %
Cashier	Healthcare and nursing assistant	CFC HCSA / Red Cross certification	12-24 months	82 %
HGV driver	Medical-supplies courier	Medical logistics certification	2 months	95%
Carpenter	Crafts teacher	Teaching diploma SFUVET / HEFP	12 months	80%
Farmer	Therapeutic gardener	Post-CFC specialisation (OrTra)	6 months	85%
Ouvrier / installateur	Networks & renewables tech.	Federal Diploma energy / photovoltaics	12-18 months	80%
Construction labourer	Energy-retrofit specialist	CFC / Minergie attestation	12-24 months	79 %

3.6 Phased Implementation

Non-negotiable sequencing principle: each phase must prove its value before the next. The surplus must exceed 2% of GDP before any propagation. v4.3.2 adds 'commons' and 'climate' milestones to each phase.

Phase	Horizon	Key actions (incl. v4.3.2)	GDP / Heures	Surplus
1: Pilot cooperative	Years 0-4	Launch of production+consumption. Carbon accounting. First local energy network pooled.	122 / 40h	0.8 %
2 : Pilote cantonal	Years 4-10	Extension to 50% of the canton. Decarbonisation Fund operational (retrofits, electrification).	166 / 37h	2.8 %
3: Start of propagation	Years 10-15	0% investment into a neighbouring canton. Cantonal commons (regional rail, grid).	213 / 35h	5.5%
4: International network	Years 15-20	Extension to 50% of cantons. 28h. Interconnected decarbonised networks.	275 / 28h	19.2%

Ecological milestones per phase (cooperative perimeter)

Phase	Net emissions (vs 2026)	Commentaire
Phase 1 (Y0-4)	baseline (0%)	Measurement set up; no significant reduction yet.
Phase 2 (Y4-10)	−30%	Decarbonisation of buildings and energy in the cooperative perimeter.
Phase 3 (Y10-15)	−50%	Mobility (rail) and regional grid decarbonised.
Phase 4 (Y15-20)	−68 %	Interconnected networks; trajectory towards net zero 2050 (CIA −65% by 2035 milestone exceeded within the perimeter).

4. Scenarios and Results

Three scenarios illustrate the model's robustness. Results at Year 20:

Indicator	Conservative	Reference	Optimiste	Base 2026
GDP index (2026 = 100)	197	275	369	100
Score GNH (/100)	63	72	74	58
Gini coefficient	0.250	0.185	0.180	0.320
Hours / week	40	28	23	41
Surplus (% GDP)	9.2%	19.2%	23.5%	0%
Cellules actives	15	20	25	0
Net emissions (vs 2026)	−45%	−68 %	−82 %	0%
Net-zero year (projection)	≈ 2055	2050	≈ 2046	—
Pooled networks	2	3	3	0

- Even the conservative scenario delivers significant progress: GDP +97%, surplus 9.2%, emissions −45%.
- The reference scenario balances all objectives and reaches net zero in 2050, in step with the CIA.
- The optimistic scenario maximises every dimension but assumes very high uptake of training and pooling.
- All scenarios maintain a surplus >2% of GDP, a necessary condition for propagation.

5. Key Assumptions

Category	Assumptions
Economic	Annual automation rate: 2 to 6% (OECD). Keynesian multiplier: 1.5. Gains captured by cooperatives. Offshore repatriation: 60% over 10 years.
Sociales	Training uptake: 25 to 80% by scenario. Cooperative coverage: 20 to 65%. Retraining acceptance: 70%. Working-time-reduction acceptance: 80%.
Politiques	40/35/25 tax key at the cantonal level. 0% propagation if surplus >2% of GDP. Democratic governance of the fund.
Communs (v4.3.2)	Progressive pooling of networks that are already mostly public (SBB, Swissgrid, Swisscom). Open access and unbundling accepted by the regulator. Capex carried by the Sovereign Fund.
Ecological (v4.3.2)	δ = 5 to 20% by scenario. Trajectory aligned with the CIA (net zero 2050). Sinks and capture not assumed mature before the end of the horizon. Imported emissions (Scope 3) included in the accounting.

- Assumptions calibrated to be consistent with the model's internal logic.
- Plausible based on the literature and precedents (Mondragon, Raiffeisen, Migros/Coop, Emilia-Romagna).
- Not empirically validated at this scale — pilot tests remain necessary.

6. Limits and Caveats

No model is perfect. v4.3.2 honestly acknowledges its limits, old and new.

Political feasibility

Requires constitutional amendments (40/35/25 key). Potential resistance from net-contributor cantons. Pooling the networks may clash with the interests of current shareholders and of cantons that own utility companies.

Economic assumptions

Baumol's cost disease — the difficulty of automating human-intensive services — is now addressed by design (§ 3.4): the model does not automate these sectors, it staffs them with the freed labour. The residual limit is that the value added of the non-automatable sector is partly non-market: it is better captured by GNH than by GDP, which may understate the real growth in well-being. In addition, diverting δ towards the climate assumes that energy costs do fall and that avoided damages do materialise.

Ecological limits (v4.3.2)

- **Hard-to-decarbonise sectors:** cement, aviation, agriculture — residual even in 2050, hence the reliance on sinks and capture that are still unproven.
- **Imported emissions:** Switzerland's real carbon footprint is largely abroad (Scope 3). Carbon accounting must include it, which tightens the target.
- **Time lag:** the model's horizon (20 years, 2026→2046) precedes the 2050 legal target; full net zero is reached beyond the reference horizon.

Limits of the commons (v4.3.2)

- **Scale:** a network has a 'natural' national — even international — scale for electricity (Switzerland is interconnected with the European grid). This pushes some decisions upward, in slight friction with strong subsidiarity (Layer 7).
- **Capture et sous-investissement :** public infrastructures are often starved of capital or captured by interests. Hence Ostrom's principles (clear rules, monitoring, separation of manager and operators) and ring-fenced capex.
- **Internet ≠ objet unique :** distinguish the physical access layer (a good national-commons candidate), the application/content layer (plural, outside any monopoly) and global connectivity (international by nature).

To address these limits

- Test in 1 to 2 pilot cantons (Geneva, Vaud) before national rollout.
- Have the assumptions independently validated by experts.
- Implement adaptively, with transparent reporting (wealth AND carbon).

7. Sources and References

Theoretical foundations

- Piketty, T. (2013). Capital in the Twenty-First Century.
- Saez, E., & Zucman, G. (2019). The Triumph of Injustice.
- Keynes, J. M. (1930). Economic Possibilities for our Grandchildren.
- Hayek, F. (1945). The Use of Knowledge in Society.
- Ostrom, E. (1990). Governing the Commons. *(foundation of Layer 11)*

Swiss data and framework

- SECO (2026) — Swiss GDP. SNB (2026) — total wealth. FSO (2026) — median wages and population.
- SERI — State Secretariat for Education, Research and Innovation. OrTra — Professional/industry organisations.
- FOEN — Climate and Innovation Act (CIA), net zero 2050, 2030/2035 milestones. *(foundation of Layer 12)*
- Network operators: SBB (rail), Swissgrid (electricity), Swisscom (telecom).

Reference websites

www.bfs.admin.ch · www.seco.admin.ch · www.snb.ch · www.bafu.admin.ch · www.bag.admin.ch · www.oecd.org · github.com/nadnone/EducationLibre (CC BY 4.0)

8. Appendices

8.1 Glossary

Term	Definition
CFC	Federal VET Certificate (Certificat Fédéral de Capacité).
RCED	Network of Democratic Economic Cells / Netzwerk Demokratischer Wirtschaftszellen.
ESS	Social and Solidarity Economy.
Gini	Inequality coefficient (0 = perfect equality, 1 = maximum inequality).
GNH	Gross National Happiness (composite well-being indicator).
SERI / OrTra	Institutions suisses de formation et organisations du monde du travail.
40/35/25 key	Tax split: 40% cantons, 35% sovereign fund, 25% equalisation.
α / β / γ / δ	Productivity-gains allocation parameters (sum = 1). δ: decarbonisation (v4.3.2).
Commun	Infrastructure governed by its stakeholders under clear, auditable rules (Ostrom); neither purely market-based nor state-run.
Unbundling	Separation of the infrastructure (common, open access) from the services running on it (plural, regulated).
LCI	Climate and Innovation Act (in force 2025); targets net zero by 2050.
Net zero	State where emissions do not exceed absorptions (neutral balance).
Secteur A / N	Automatable (A) vs non-automatable (N) sector. Labour freed from A is redirected to N (§ 3.4).
Freed labour	Workers whose job is automated, retrained (Layer 8) into the non-automatable sector rather than laid off.
κ (kappa)	Annual net capture rate of GDP into the Sovereign Fund (≈ 1.16%/yr in the reference case).

8.2 Mathematical Formulas

1. Gains-allocation constraint (extended v4.3.2)

```
α + β + γ + δ = 1
```

2. Productivity gain (recurring annual rate)

```
ΔP(t) = r          (r = annual automation rate, 2 to 6%)
```

3. Working-time reduction

```
H(t) = H(t-1) × (1 − α × ΔP(t))
```

4. Cumulative Sovereign Fund surplus

```
S(t) = S(t-1)
     + β × ΔP(t) × GDP(t) × fiscal_redistribution × 0.35
     + D_wages(t)
```

v4.3.2 recalibration: the simplified implementation of this formula underestimated the surplus at 7.7% of GDP. The Fund is now calibrated on a net capture rate κ ≈ 1.16%/yr of GDP (40/35/25 fiscal channel, β productivity channel, D_wages and green revenues, net of Fund expenditure), which reproduces the published surplus of 19.2% of GDP at Y20.

5. Sectoral reallocation

```
R(t) = γ × ΔP(t) × L(t)
```

6. Decarbonisation Fund *(nouveau v4.3.2)*

```
F_decarb(t) = F_decarb(t-1)
            + δ × ΔP(t) × GDP(t) × fiscal_redistribution × 0.10
            + green_revenues(t)
```

Definition of the 0.10 coefficient: this is the share of fiscal redistribution earmarked for decarbonisation (10%), strictly parallel to the 0.35 coefficient earmarked for the Sovereign Fund in formula 4. The two earmarks (0.35 'fund' + 0.10 'decarbonisation') are internal allocations of the redistributed flow and do not change the cantonal 40/35/25 key, which remains the decomposition of the captured base. The 0.10 coefficient deliberately echoes the level of δ but applies here to the fiscal channel, distinct from the productivity-gains channel (δ).

7. Bilan carbone net *(nouveau v4.3.2)*

```
E_net(t) = E_brut(t) − A_puits(t) − C_captage(t)
E_brut(t) = E_brut(t-1) × (1 − ρ(t))
Cible : E_net(2050) ≤ 0
```

8. GDP decomposition *(v4.3.1)*

```
Y_trav(T) / Y_trav(0) = P_auto × P_coop × ( H(T) / H(0) )
  P_auto = (1 + r)^T          (productivity/hour, automation)
  P_coop = cooperative-extension factor
Reference (T=20): 2.19 × 1.84 × 0.68 ≈ 2.75   (+175%)
```

9. Two-sector dynamics *(nouveau v4.3.2)*

```
L(t) = L_A(t) + L_N(t)                    (effectif total constant)
L_A(t) = L_A(t-1) × (1 − a)               (a = displacement by automation)
F(t)  = L_A(t-1) × a                      (freed labour)
L_N(t) = L_N(t-1) + g × F(t)              (g = share redirected to N)
Reference: g = 1 (100% reallocated); a = min(sector_rate, r)
```

10. Non-automatable sector capacity *(nouveau v4.3.2)*

```
Q_N(t) = Q_N(t-1) × ( L_N(t) / L_N(t-1) )    (Baumol: capacity ∝
headcount)
Reference (T=20): Q_N ≈ 162   (+62% care/education capacity)
```

8.3 Sensitivity Analysis

Integrating the mainstream education system stabilises the model (retention through the value of qualifications). The δ parameter explicitly trades off decarbonisation speed against short-term growth.

Assumption	Base case	Tested case	GDP impact (Y20)	Gini impact (Y20)
Automation rate	4%/yr	2%/yr	−50 pts	+0.02
Automation rate	4%/yr	6%/yr	+100 pts	−0.01
Cooperative coverage	45%	20%	−80 pts	+0.04
Cooperative coverage	45%	65%	+90 pts	−0.02
Fiscal redistribution	28%	20%	−40 pts	+0.03

Assumption	Base case	Tested case	GDP impact (Y20)	Gini impact (Y20)
Fiscal redistribution	28%	35%	+50 pts	−0.02
α (time reduction)	45%	30%	+18 pts	0.00
β (Fonds Souverain)	35%	30%	−12 pts	+0.01
γ (reallocation)	10%	20%	−10 pts	+0.01

Ecological sensitivity (v4.3.2)

Assumption	Base case	Tested case	GDP impact (Y20)	Emissions (Y20)
δ (decarbonisation)	10%	5%	+5 pts	+12 pts (slower)
δ (decarbonisation)	10%	20%	−8 pts	−15 pts (faster)
Decarbonisation pace ρ	medium	high	−3 pts	−18 pts
Pooled networks	3	1	−15 pts	+20 pts (rent + costs)

Contrepartie (contrainte $\alpha + \beta + \gamma + \delta = 1$): in these tests, changes in δ are taken from β (Sovereign Fund), with α and γ held constant. This is why a higher δ (hence a lower β) costs a few points of short-term GDP — less capital reinvested through the multiplier — while accelerating decarbonisation. Pooling the networks, by contrast, works both ways: it accelerates decarbonisation AND protects GDP from rent extraction.